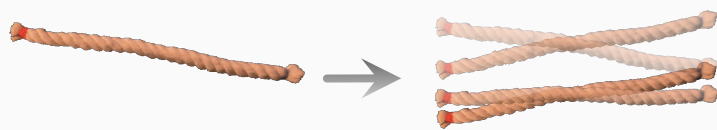


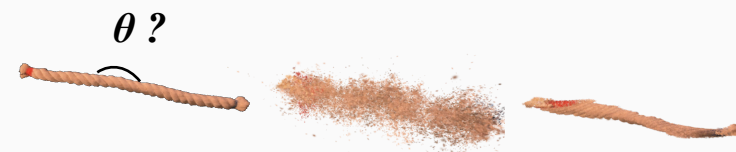
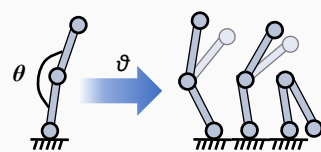
DeformGen

(a). State Challenge: High-DoF & Dynamic Constraint

Only Spatial Diversity $\times$



NO Explicit Kinematic Params θ $\times$



Dynamic Topological Transformation



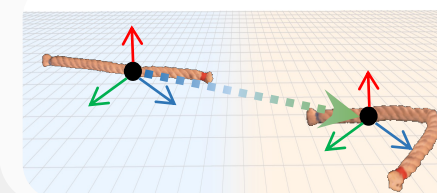
Topological Diversity $\checkmark$



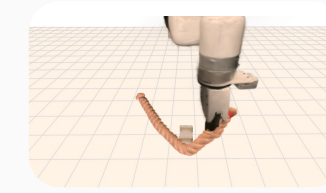
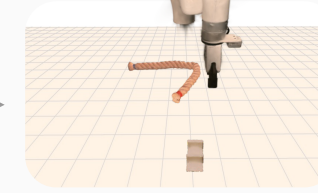
Realistic States via Physical Rollouts $\checkmark$

(b). Trajectory Challenge: Non-Equivariance

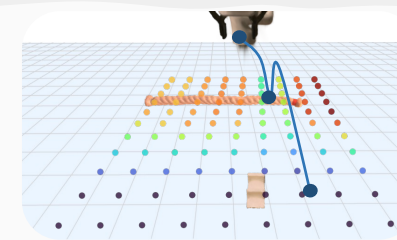
Contact Mismatch $\times$



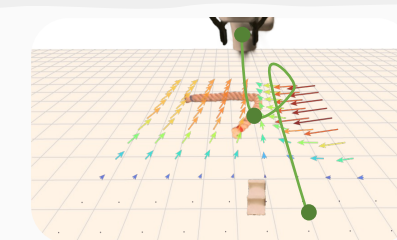
NO Compensation $\times$



Deformation Field Warp

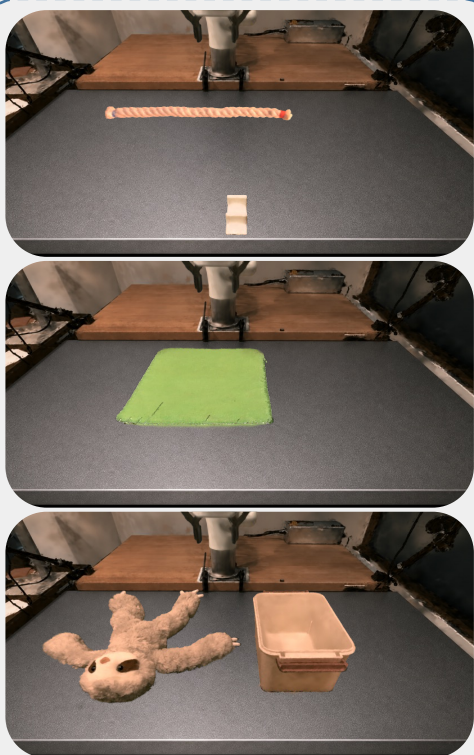


Contact Capture $\checkmark$

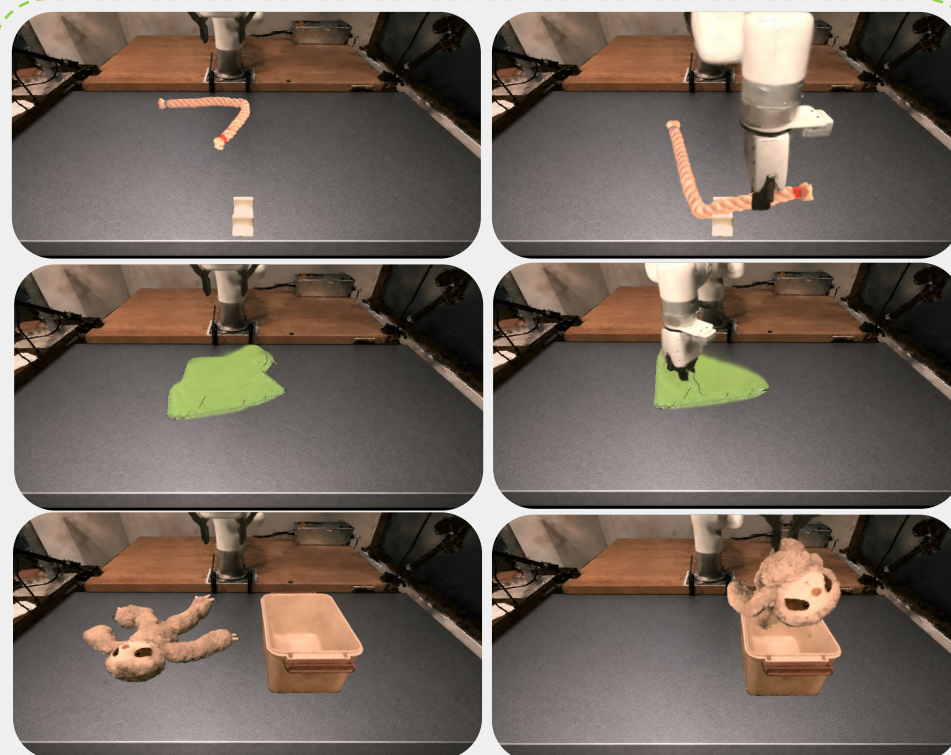


Trajectory Compensation $\checkmark$

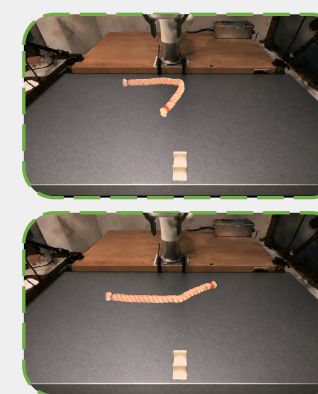
One Demonstration



Augmented Data



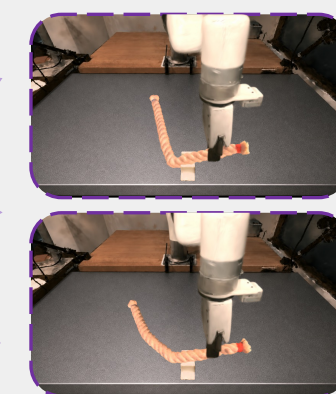
(c). Data Synthesis via Topological Augmentation



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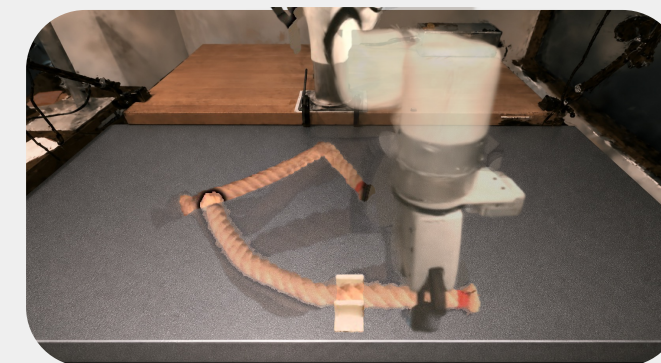


Policy

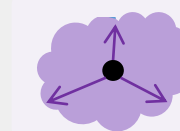


...

Generalize to Unseen States



Higher Success Rate



Topological Generalization

(d). Policy Training & Evaluation